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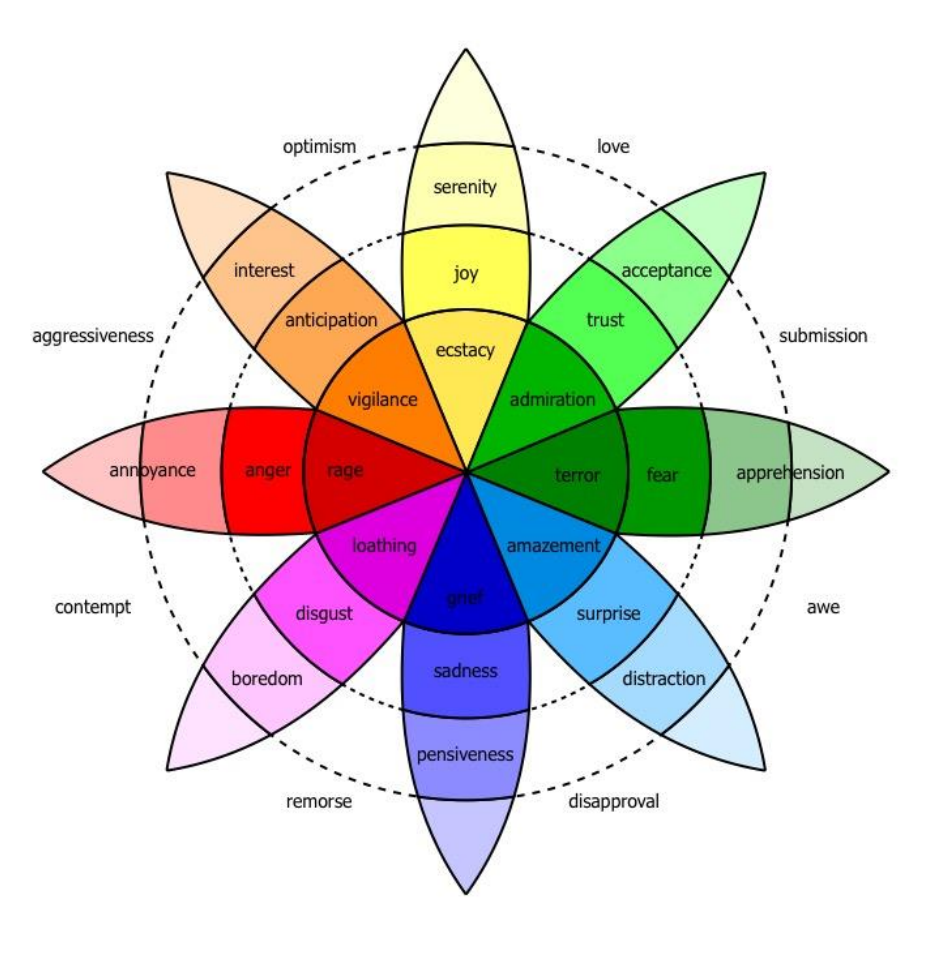
SENTIMENT ANALYSIS

DECODING EMOTIONS: EXPLORING SENTIMENT THROUGH LEXICON ANALYSIS



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Share your opinion!

What can news and social media reveal about people's emotions during and about war?

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Agenda

Introduction

Sentiment Analysis (SA)

Methods

Theory of Emotions

Lexicon

Sentiment Lexicons

Resources

Sentiment Analysis in Python

Python Library

Implementation

Consideration and Future

Challenges

Ethical considerations

Future of SA

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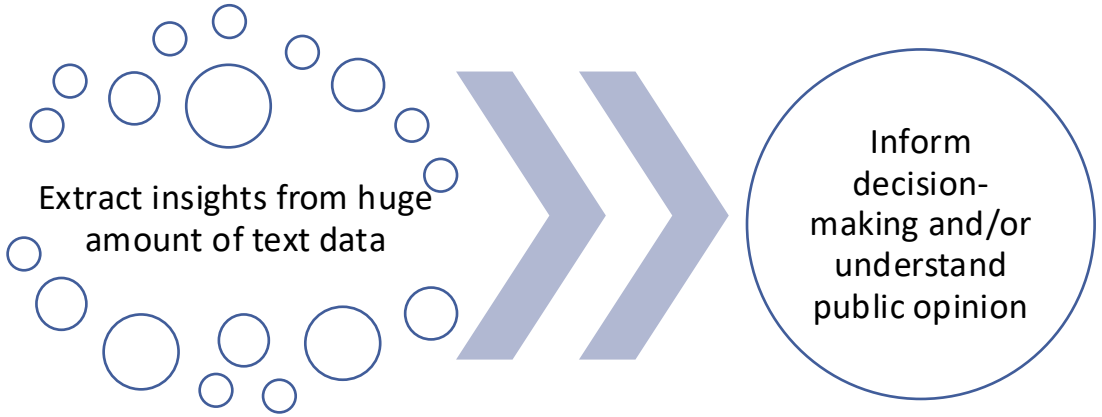
Sentiment Analysis

What is Sentiment Analysis?

- A technique used to find the emotional tone or sentiment in a piece of text.
  - Positive, Negative, Neutral.
  - More complex emotions: Happiness, Anger, Disappointment.
- Uses NLP techniques to understand the context, tone, and polarity of words and phrases.



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Role of NLP in Sentiment Analysis

Text Understanding	Cleaning and preparing the text data for analysis.
Feature Extraction	Convert text into a numeric (BoW or TF-IDF).
Sentiment Classification	Classify the sentiment of the text or detect specific emotions
Context and Nuance Understanding	DL models can grasp the context and nuances of language, such as sarcasm or irony.
Continuous Learning	Can improve over time through ML techniques, adapting to new slang, jargon etc.

Real Life Application

Why Sentiment Analysis?

Business & Marketing	Improve products, and change marketing strategies
Social Media Monitoring	Collect public sentiment on different topics, brands, or events
Customer Service	Categorize customer feedback into complaints and compliments
Finance & Markets	Gain insights into market trends for informed investment decisions
Politics & Public Policy	Collect public opinion on policies, campaigns, and social issues

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Methods

How to do Sentiment Analysis?

Rule-based Method

Relies on predefined rules - also known as lexicons

Identifies sentiment-bearing words and analyzes context

Use logical operations (sum, majority voting, threshold value) to get sentiment polarity

Machine Learning Method

Training a model on labelled data (in large amounts)

Learns patterns and relationships between features and sentiment labels

Utilizes supervised learning algorithms like SVM or neural networks

Rule based vs. ML

Feature	Rule-Based Systems	Machine Learning Approaches
Methodology	Uses predefined sets of rules and lexicons	Learns from data, using statistical models
Example	VADER and SentiWordNet	SVM or neural networks based
Adaptability	Limited - requires manual updates to rules	High - models adapt by learning from new data
Data Requirement	Low - does not require training data	High - requires large datasets for training
Context Sensitivity	Limited - struggles with context expressions	High - better at understanding context
Language Evolution	Slow to adapt to new slang, jargon	Quickly adapts as it learns from current data
Application Scenarios	Effective for simple, clear tasks	Suited for complex, nuanced texts
Precision and Recall	May vary – rule specific	Generally higher- with sufficient training

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Which of the following method do you think is more effective and why? (1 minute)

1. **Rule based**
2. **ML based**

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Two Theories of Emotions

Theory 1. Atomic basic emotions

A list of 6 or 8 basic emotions

Others are generated from these basic emotions.

E.g. Ekman's six basic emotions (6)

E.g. Plutchik's wheel of emotions (8)

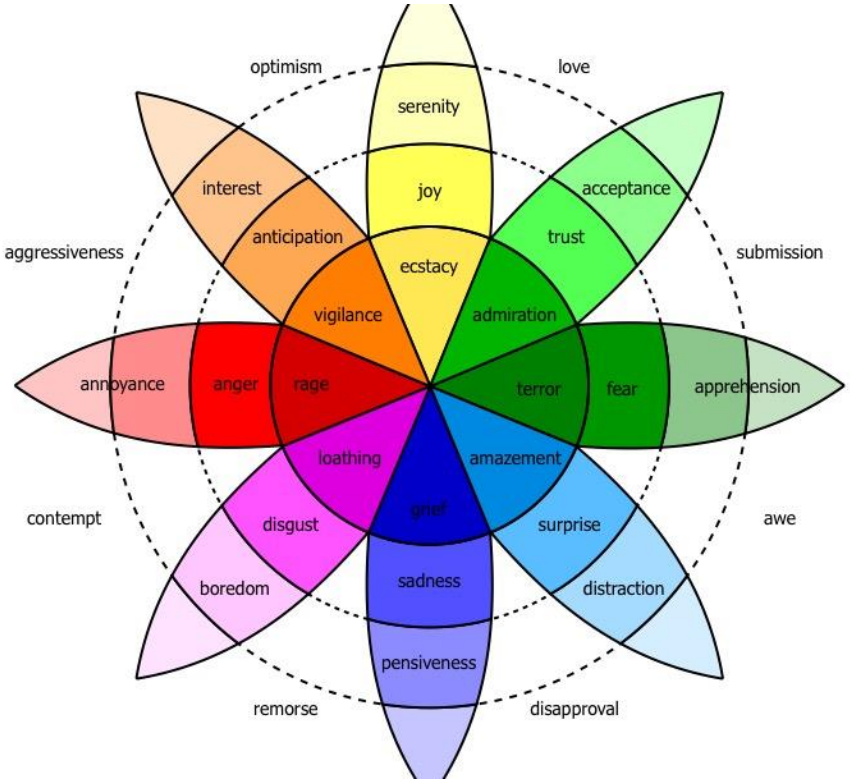
Theory 2. Emotion is a point in two- or three-dimensional space

Valence: tells the positivity or negativity of an emotion

Arousal: tells the intensity associated with an emotional experience.

Control or Dominance: tells the degree to which an emotion feels under the individual's control or influence

Theory 1. Atomic Basic Emotions



Plutchik's wheel of emotions

Ekman's six basic emotions

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Theory 2. Valence-Arousal Dimensions

Picture Source: https://aclanthology.org/N16-1066/

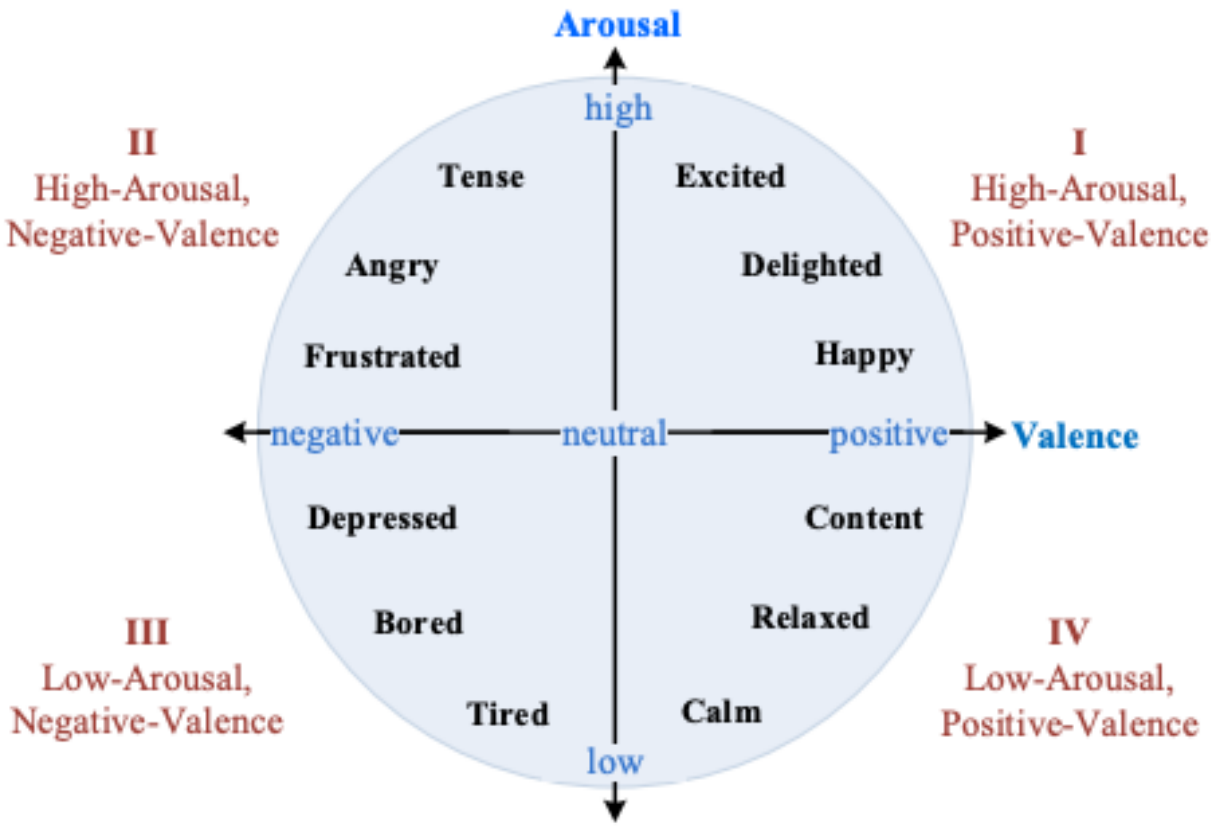

Figure 1: Two-dimensional valence-arousal space.

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How can we decipher emotions from written text using predefined word lists? (2 minute)

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What are Lexicons?

A lexicon is a complete set of words in a language, including their

Meanings

Usage

Pronunciation

A structured collection of words or phrases with associated linguistic information (NLP)

Part-of-speech tags

Semantic categories

Sentiment scores

Usage of Lexicons in NLP

Word sense disambiguation

Sentiment analysis

Named entity recognition

Sentiment lexicons

Emotion lexicons or Affect lexicons

Dictionaries containing words or phrases annotated with their associated emotional categories.

Tool for identifying the sentiment polarity of words and texts.

Words are categorize based on their sentiment polarity

(positive, negative, neutral).

May contain the numerical scores representing the intensity of the sentiment.

WordNet - a lexical database of English words

Nouns, verbs, adjectives and adverbs are grouped into sets of cognitive synonyms (synsets), each expressing a distinct concept.

Synsets are interlinked by means of conceptual-semantic and lexical relations.

Resources: Sentiment Lexicons

AFINN (Affective Norms for English Words) Developed by Finn Årup Nielsen

- A list of 3300+ words ranked according to their sentiment polarity between -5 to +5.
- Quick Sentiment Estimation
- Content Filtering
- Prototype Development

VADER (Valence Aware Dictionary and sEntiment Reasoner)

- A rule-based pre-trained model designed for social media contexts.
- Uses a valence score (-1 to +1) for each word to determine its positivity or negativity.
- Intensity of the sentiment, can be determined by capitalization and punctuation.
  - Social Media Monitoring
  - Market Analysis and Customer Feedback

Resources: Sentiment Lexicons

SentiWordNet

- Assign three sentiment scores: (positivity, negativity, and objectivity) to each synset (groups of synonymous words) of WordNet.
- To see the full paper click here

LIWC (Linguistic Inquiry and Word Count)

- A text analysis program that categorizes words into various psychological, emotional, and linguistic categories. (Link)
- Used to understand psychological states and social behaviors.

Python Library and Tools

NLTK

Offers lexical resources such as **WordNet and VADER** tool.

TextBlob

Uses a lexicon of pre-scored words to estimate the polarity and subjectivity of a text.

Tensor-Flow

Open-source machine learning framework by Google.

Used to build and train DL models to classify text into different sentiment categories.

Powerful for understanding the context and nuances in text data.

PyTorch

Open-source machine learning library by Facebook.

Like TensorFlow, it supports RNNs, CNNs, and transformers

Popular due to its flexibility and dynamic computational graph.

Python Library and Tools

Feature/Tool	NLTK	TextBlob	TensorFlow	PyTorch
Core Strength	Comprehensive NLP toolkit	Simplicity and ease of use	Scalable deep learning	Flexible and dynamic modeling
Ease of Use	Good, with some learning curve	Excellent for beginners	Steep learning curve	Intuitive for developers
Performance	Good for educational/research	Good for small-scale projects	High, suited for production	High, with dynamic computation
Best Suited For	Educational/research projects	Quick prototyping, beginners	Large-scale, production projects	Research, custom models
Sentiment Analysis	Lexicon-based (VADER)	Built-in, lexicon-based	Custom deep learning models	Custom deep learning models
Model Flexibility	Limited by available tools	Limited to built-in features	Highly customizable	Highly customizable
Community Support	Large, especially in academia	Moderate	Very large	Large, especially in research
Deployment	Not specifically designed for production	Not specifically designed for production	Robust tools for deployment	Growing support for deployment

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Python Implementation

Scoring in lexicon-based sentiment analysis

Prepare a Sentiment Lexicon

Use a lexicon where words are tagged with sentiment scores.

Scoring can be binary, scale-based or categorical.

Text Processing

Clean and tokenize the text into words.

Match & Score

For each word in the text, find its sentiment score in the lexicon.

If word is not in the lexicon either ignore or assign a neutral score.

Aggregate

Sum the scores of all words to get an overall sentiment score for the text.

Interpretation

Interpret the aggregate score to determine the text's overall sentiment.

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Why is understanding context challenging in sentiment analysis? (2 minute)

Discuss in pair and share the summary here

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Challenges

Context dependency

- The sentiment value of words can vary greatly depending on the context.
- Ex: Sent 1: The device's performance is unpredictable.
- Sent 2: The plot is delightfully unpredictable.
- "unpredictable" has a negative sentiment in Sent 1 but positive in Sent 2.

Sarcasm

- Detecting sarcasm requires understanding broader context and speaker's intent.
- Ex: I just love getting stuck in traffic.

Challenges

Ambiguity

A word, phrase, or sentence can have multiple interpretations.

E.g. "light" can refer to weight (positive for laptops) or intensity (negative for meal).

Cultural Differences

A lexicon developed for one cultural or linguistic context might not be applicable or accurate in another

E.g. He's the Michael Jordan of coding. (Implying greatness)

Other Challenges

Syntax and Structure	• I don't hate it. (Negation changes sentiment)
Domain-Specific Language	• The system experienced a crash. (Negative in everyday use but could be neutral in tech discussion)
Long-Range Dependencies	• After all the terrible things that happened, today was actually good. (Earlier context needed to understand final sentiment)
Slang	• This new phone is sick! (Meaning excellent)
Emotional Complexity	• I'm happy you're leaving but sad to see you go.

Ethical Considerations

Bias in Lexicons

Might include biases from their creators, like cultural biases or showing some sentiments more than others.

Leads to biased results.

Misinterpretation and Misuse

Ignoring context, sarcasm, ambiguity, or cultural differences can lead to wrong sentiment analysis.

This could misunderstand what people think or feel, unfairly affecting decisions.

Transparency and Accountability

Important to know “How lexicons are made and used in sentiment analysis”.

This helps people understand why and how sentiments are being studied.

Future of Lexicon-Based Sentiment Analysis

Dynamic Lexicon Updates

Automatically add new slang and expressions from the internet.

Update continuously based on current trends and how people are using language globally.

Adjust based on how people are using words in different situations.

Machine Learning Integration

Look for hybrid models that combine lexicon accuracy with deep learning's contextual understanding.

Interactive Systems

Anticipate systems that learn from user feedback to refine accuracy

Cross-lingual Capabilities

Enhance multilingual sentiment analysis, using advanced ML techniques to apply insights across languages.

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What is the biggest challenge in lexicon-based sentiment analysis? (2 minute)

Discuss in pair and share the summary here

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Key Takeaways

Principle
Analyzes sentiment by summing up scores of words from a lexicon.

Advantages
Simple, no training data needed, transparent analysis.

Limitations
Struggles with context, fixed word sentiments, language evolution.

Improvements
Use of contextual rules and domain-specific lexicons.

Hybrid Approaches
Combines with machine learning for nuanced analysis.

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Feedback-06

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